Prepare DP1 paper (SP-2021)

[SP-2710] Respond to review feedback on Section 3: Data Products

Created: 24/Oct/25 Updated: 30/Jan/26 Due: 14/Nov/25

Resolved: 30/Jan/26

Status:	Done
Project:	System Performance
Components:	paper
Fix versions:	None
Parent:	Prepare DP1 paper

Type:	Story	**Priority:**	Undefined
Reporter:	Leanne Guy	**Assignee:**	James Mullaney
Resolution:	Done	**Votes:**	0
Labels:	DP1		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original estimate:	Not Specified		

Reviewers:	Leanne Guy
Sprint:	
RubinTeam:	Verification & Validation

Description

Respond to review comments on Section 3:

Comments

Comment by Leanne Guy [24/Oct/25]

Additional “minor comments” to address;

- Line 694 times --> \times
- The description of the CcdVisit catalog on page 16 should be after the discussion of the SSObject and SSSource catalogs so that all detection catalogs are listed together.

Comment by Leanne Guy [24/Oct/25]

There's significant description about data release processing, but overall there's little description of how the images were reduced (bias subtracted, flat-fielded, cosmic rays identified, sources extracted, image subtraction) in terms of the algorithms and software. Although this information may be available in other online resources, technotes, and submitted or soon to be submitted papers (which should be signposted in this manuscript), the reader should be able to have a general understanding of the Rubin pipelines in the state they were in when DP1 was processed without having to read all of the contents of the software pipeline papers.

A figure similar to Figure 5 of Aihara et al (2018; https://ui.adsabs.harvard.edu/abs/2018PASJ...70S...8A/abstract) to provide a flow chart of the data processing stages and data products generated at each stage would be helpful.
Section 3:

Patches and tracts are not well explained in terms of the skymap and how the sky skymap is produced. I could do the calculation, but particularly for a new reader it's not clear to whether the tracts/patches used in DP1 are the same size as what will be used for LSSTCam or is the grid different. Are patches in this case the same size as an individual LSSTCam/ComCam CCD detector? Please give the value for the overlap between neighboring patches and neighboring tracts. A diagram would help to explain this.

Line 701 - "The size of a tract is larger than the LSSTCam field of view; however, since each observed field extends across more than one tract, each field covers multiple tracts." Table 3 Caption: The size of a tract is larger than the LSSTCam field of view; however, since each observed field extends across more than one tract, each field covers multiple tracts." LSSTCam's field of view is 9.6 deg^2. ComCam covers 0.5 deg^2 . Tracts cover 2.8 deg^2. Sould this be ComCam instead of LSST in these sentences?

Is all data processing/source identification performed on a patch/tract level or just the co-adding step? Is image subtraction performed on a patch or track level? From Reading Subaru HSC data release papers and DESC (Dark Energy Science Collaboration) DC2 paper (DESC 2021; https://ui.adsabs.harvard.edu/abs/2021ApJS..253...31L/abstract), it seems like difference imaging template creation and subtraction is performed on a patch/tract level.

Table 3 may be useful but not knowing the tessellation ID grid for tracts, it's not clear to me a reviewer/reader as how important this information is. Is there anything at DP1 data products level that a data rights holder can access with these tract IDs? Table 3 may be better suited to an Appendix as most DP1 users will not likely need this information, though useful to have for those deep diving into the data processing.

Table 4 is not referenced in the paper.

Line 660 - what are CTI models and brighter-fatter kernels?

It would be helpful to identify if the 153 raw images that failed processing are scattered across all of the fields or in only certain fields. From Lines 771-773 it appears that the low galactic latitude Seagull and 47_Tuc contain most of these failed raw images.
template co-adds - what criteria was used to make templates. What were the seeing requirements per filter? How many images per patch went into a template.

Footnote 88 should be in the main text.

Section 3.2 - It would be helpful to have in some table form the number of sources and objects, diaObjects, diaSources, and SSsources, SSobjects per field in total and per filter

Section 3.2 - Early on the section, there should be a clear link to the data schema website for DP1 https://sdm-schemas.lsst.io/dp1.html You may want to consider moving parts section 6.5 to earlier sections in the paper.

Section 3.2 - I do recommend that the authors consider also adding in the data schema for the various tables in DP1 as supplemental tables in the appendix given websites can disappear

Section 3.2.1 - How many rows are in the SSObject catalog?

Section 3.3 - Line 1239 "Many more survey property maps are available in the DP1 repository"- Without going to the documentation, the reader should be able to determine from this overview paper what maps are available. Please provide a table in the Appendix with the list of all property maps that are available for DP1.

Section 3.4 - Can you comment further about how the deep coadds in the various filters are combining together?

Section 3.5 -Visits table and CcDVisit catalog? The CcdVisit table is called a catalog in Section 3.2, but the Visits table is called a table in the Metadata section. Section 3.5 doesn't mention the CcdVisit table at all. I find this confusing. If I'm looking for the conditions/properties of a specific DP1 observation, I likely want a combination of data queried from the CcDVisit/Visits tables. I imagine many people will skim the paper and find the metadata section missing that the mention of CcDvisits table/catalog. In both

Sections 3.2 and 3.5 clarify where all the useful data about an observation taken is stored.

Section 3/Section 4 - It would be helpful to have normalized histograms per filter showing the magnitude distribution (flux distribution) of detected DP1 sources. One plot for the coadd source catalogs and one for the single epoch source catalogs.
Limiting Magnitude/Depth - There's little discussion of the limiting magnitude/depth of this dataset particularly single epoch image processing/processed visit images (PVIs). I see there is a 5-sigma limiting magnitude in the CcdVisit table. There should be a histogram/visualization of the limiting magnitudes per filter to give the reader a sense of the depth of the

Comment by James Mullaney [05/Nov/25]

>> There's significant description about data release processing, but overall there's little description of how the images were reduced (bias subtracted, flat-fielded, cosmic rays identified, sources extracted, image subtraction) in terms of the algorithms and software. Although this information may be available in other online resources, technotes, and submitted or soon to be submitted papers (which should be signposted in this manuscript), the

reader should be able to have a general understanding of the Rubin pipelines in the state they were in when DP1 was processed without having to read all of the contents of the software pipeline papers.

When planning the paper, we made the conscious decision to describe the data products prior to processing pipelines, as we understand that most readers will be more interested in the data products over the detailed description of the processing steps. We stick by this decision, but this query highlights a good point in that it should be made clear at the outset of Section 3 that its focus is data products alone, and that the descriptions of the pipelines and means of access are provided in later sections. The following has been added to the first paragraph of Section 3:
”To aid legibility, we have separated the descriptions of the data products from the description of the data release processing pipeline, a summary of which is provided in S4.”

Similarly, because most of the data products described here can be accessed via either the \gls
{IVOA}

Services (S6.2.1) or the Data \gls
{Butler}

(S6.2.2), we describe them in a manner that is agnostic to the means of access.

>> A figure similar to Figure 5 of Aihara et al (2018) to provide a flow chart of the data processing stages and data products generated at each stage would be helpful.

Similarly, I think this should appear in Section 4.

>> Patches and tracts are not well explained in terms of the skymap and how the sky skymap is produced. I could do the calculation, but particularly for a new reader it's not clear to whether the tracts/patches used in DP1 are the same size as what will be used for LSSTCam or is the grid different. Are patches in this case the same size as an individual LSSTCam/ComCam CCD detector? Please give the value for the overlap between neighboring patches and neighboring tracts. A diagram would help to explain this.

We have added the following at the start of Section 3:
”To facilitate processing, Rubin DP1 uses a single skymap* that covers the entire sky area encompassing the seven DP1 fields. The DP1 skymap divides the entire celestial sphere into 18938 tracts, each covering approximately 2.8 deg^2. The tract are arranged in rings of declination, ordered from south to north, then with increasing right ascension within a ring. Each tract is further subdivided into 10$\times$10 equally-sized patches. Both tracts and patches overlap with their neighboring regions. The amount of overlap between tracts changes with declination, with tracts nearest the poles having the greatest degree of overlap; the minimum overlap between tracts is 1 arcminute. By contrast, the amount of overlap between patches is constant, with each patch overlapping each of its neighbouring patches by 80 arcseconds. As a consequence, each patch covers, 0.036 deg^2, which is slightly larger than the tract area divided by the number of patches in a tract. The aerial coverage of a patch is comparable to, but somewhat smaller than, the 0.051 deg^2 field-of-view of a single LSSTComCam or LSSTCam detector, meaning each detector image spans multiple patches.
*A skymap is a tiling of the celestial sphere, organizing large-scale sky coverage into manageable sections for processing and analysis. While the skymap described here is specific to DP1, we do not anticipate major changes to the skymap in future data releases.

>>> Line 701 - "The size of a tract is larger than the LSSTCam field of view; however, since each observed field extends across more than one tract, each field covers multiple tracts." Table 3 Caption: The size of a tract is larger than the LSSTCam field of view; however, since each observed field extends across more than one tract, each field covers multiple tracts." LSSTCam's field of view is 9.6 deg^2. ComCam covers 0.5 deg^2 . Tracts cover 2.8 deg^2. Sould this be ComCam instead of LSST in these sentences?

Yes - thank you. Changed to LSSTComCam.

>> Is all data processing/source identification performed on a patch/tract level or just the co-adding step? Is image subtraction performed on a patch or track level? From Reading Subaru HSC data release papers and DESC (Dark Energy Science Collaboration) DC2 paper (DESC 2021; https://ui.adsabs.harvard.edu/abs/2021ApJS..253...31L/abstract), it seems like difference imaging template creation and subtraction is performed on a patch/tract level.

Early processing steps, such as instrument signature removal (e.g., bias-subtraction, flat-fielding, etc) and the calibration of images from a single detector are performed at the per-visit-per-detector level. Later processing steps which involve the production, analysis, and utilisation of coadds are performed at patch and tract level. It is explained at the start of Section 3 that the skymap is integral to the production of coadds, and that coadd production is done at the patch level. However, we have strived to avoid using terms like “visit-level” or “tract-level” in Section 3, as these are largely Butler-specific terms, whereas Section 3 aims to be agnostic to means of accessing the data (e.g., the user does not need to know that the object table is tract-level if they are accessing it using the PAT protocol). Instead, we refer the reader to Section 6 to learn how to access the data via the various means:

“Data products are accessed via either the IVOA Services (S6.2.1) or the Data Butler (S6.2.2).”

In light of this, we have removed reference to “visit_level” and “tract-level” in Section 3, although we maintain the description of tracts and patches as we need to refer to them when describing the aerial coverage of a coadd.

On the specific point about image subtraction: Difference images are created at the visit level, but the templates that are subtracted are generated from patch template coadds which are stored at the patch level. We feel the statement “In total, there are 15972 difference_images in DP1, one for each visit_image….contain the same number of pixels, which is the same as that of the input visit_image.“ in Section 3 addresses that point.

>> Table 3 may be useful but not knowing the tessellation ID grid for tracts, it's not clear to me a reviewer/reader as how important this information is. Is there anything at DP1 data products level that a data rights holder can access with these tract IDs? Table 3 may be better suited to an Appendix as most DP1 users will not likely need this information, though useful to have for those deep diving into the data processing.

The tract ID is the primary means of referring to and therefore accessing coadded images and object tables within the butler environment. As such, we feel that providing this information within the main body of the paper is useful to the reader. We agree, however, that this information should be presented in Section 6 at the point where we describe data coordinates within the butler environment.

>> Table 4 is not referenced in the paper.

Table 4 is referenced in the paper where we describe raw images. The word “Table” is split across two lines, so may not have shown-up in a search.

>> Line 660 - what are CTI models and brighter-fatter kernels?

References to CTI and brighter-fatter kernels have been removed from this highest-level list. Rather than describe them here, we have instead added forward references to the sections in which each type is covered.

>> It would be helpful to identify if the 153 raw images that failed processing are scattered across all of the fields or in only certain fields. From Lines 771-773 it appears that the low galactic latitude Seagull and 47_Tuc contain most of these failed raw images.

As anticipated, most of these (131) are for 47_Tuc, but none are for Seagull. The others are Rubin_SV_095_-25: 9, ECDFS: 8, EDFS_comcam: 2, Fornax_dSph: 3. These numbers are now included in the text:
”The majority of failures – 131 in total – were due to challenges with astrometric fits or PSF models in the 47_Tuc crowded field. The other failures were in the Rubin_SV_095_-25 (9 failures), ECDFS (8), Fornax_dSph (3), and EDFS_comcam (2) fields.”

>> template co-adds - what criteria was used to make templates. What were the seeing requirements per filter? How many images per patch went into a template.

The criteria is explained in the text. It is largely based on PSF size, but there isn’t a specific cutoff:
”In the case of template_coadds, one third of visit_images covering the patch in question with the narrowest PSF FWHM are selected. If one third corresponds to fewer than twelve visit_images (i.e., there are fewer than 36 visit_images covering the patch), then the twelve visit_images with the narrowest PSF FWHM are selected. Finally, if there are fewer than twelve visit_images covering the patch, then all visit_images are selected.”

The number of images per patch varies substantially across the survey, and is not consistent across a given patch; one part of a patch could be covered by many more visits than another part of the same patch, especially toward the edges of a field. As such, this is not a straightfoward number to report and may even be misleading. The following sentences have been added to explain as such:

“The number of visit_images that contributed to a given deep_coadd varies across the patch; the Survey Property Maps can be consulted to gain insights into the total exposure time at all locations covered by the survey.”

and

“As with deep_coadds, the number of visit_images that contributed to a template_coadd varies across the patch.“

>> Footnote 88 should be in the main text.

Done, although we have retained the following in a footnote:
”For storage space reasons, warped templates are not retained for \gls
{DP1}

, as they can be readily and reliably recreated from the \texttt
{template_coadd}

s.”

>> Section 3.2 - It would be helpful to have in some table form the number of sources and objects, diaObjects, diaSources, and SSsources, SSobjects per field in total and per filter

After some discussion, we have decided not to do this. Users will be able to obtain these numbers from the Rubin Science Platform, should they wish. Further, reporting such numbers will become unsustainable with future data releases, and we don't wish to set such precedents with this first Data Preview paper.

>> Section 3.2 - Early on the section, there should be a clear link to the data schema website for DP1 https://sdm-schemas.lsst.io/dp1.html

The first sentence of section 3 now reads:
Here we describe Rubin DP1 data products and provide summary statistics for each, but we also refer the reader to the DOI-registered DP1 release documentation available at \url
{https://dp1.lsst.io}

and the catalog schemas available at \url
{https://sdm-schemas.lsst.io}

.\footnote{Searchable catalog schemas are also available to Data Rights Holders via the Rubin Science Platform at \url
{https://data.lsst.cloud}

.}

>> You may want to consider moving parts section 6.5 to earlier sections in the paper.

This was discussed at length during the planning stages of the paper, and it was decided that since there are multiple means by which to access the data, the descriptions of the data products should be decoupled from the descriptions of how the data is accessed. We feel that frequently referring to variously IVOA-specific and Butler-specific concepts in Section 3 would lead to confusion. A sentence directing the reader to Section 6 for information regardinng data access has been added to the first paragraph of this section.

>> Section 3.2 - I do recommend that the authors consider also adding in the data schema for the various tables in DP1 as supplemental tables in the appendix given websites can disappear.

While we understand the referee's concerns, we don't feel that this will be of substantial benefit. We note that the DP1 release web pages are DOI registered, and links to the schemas for all catalogs included in DP1 are included in those pages. Further, the schemas are readily available via the Rubin Science Pipeline, so as long as the data is available, then users have full access to the schemas. The text footnote at the start of the Section 3 points this out.

>> Section 3.2.1 - How many rows are in the SSObject catalog?

The following has been added the the SSObject section:
"The SSObject catalog contains data for 431 SSObjects in DP1."

>> Section 3.3 - Line 1239 "Many more survey property maps are available in the DP1 repository"- Without going to the documentation, the reader should be able to determine from this overview paper what maps are available. Please provide a table in the Appendix with the list of all property maps that are available for DP1.

A summary of all the Survey Property Maps is included in the relevant section:
”The available maps describe total exposure times, observation epochs (one each for the earliest, mean, and latest observation epoch), PSF size and shape (one for each of the e^1 and e^2 shape parameters; see \secref
{ssec:psf_models}

), PSF magnitude limits, sky background and noise levels, as well as astrometric shifts (one each for right ascention and declination) and PSF distortions (one for each of the e^1 and e^2 shape parameters) due to wavelength-dependent atmospheric DCR effects.”

We have deleted the reference to “many more Survey Property Maps are available” as they are now all listed.

>> Section 3.4 - Can you comment further about how the deep coadds in the various filters are combining together?

We have added a reference to Lupton and O’Mullane (2015), which describes the process used to combine multi-band images:

DP1 provides multi-band HiPS images created by combining data from individual bands of deep_coadd and template_coadd images, using an improved version (Lust et al. in prep) of the algorithm presented in Lupton et al (2004).

>> Section 3.5 -Visits table and CcDVisit catalog? The CcdVisit table is called a catalog in Section 3.2, but the Visits table is called a table in the Metadata section. Section 3.5 doesn't mention the CcdVisit table at all. I find this confusing. If I'm looking for the conditions/properties of a specific DP1 observation, I likely want a combination of data queried from the CcDVisit/Visits tables. I imagine many people will skim the paper and find the metadata section missing that the mention of CcDvisits table/catalog.

The following has been included in the description of the CCDVisit catalog:

“When combined with the data contained in the \texttt
{Visit} table described in \secref{sec:metadata}, it provides a full picture of the telescope pointing and sky conditions at the time of observation.”

and the following has been included in the description of the Visit table:
”DP1 also includes metadata about the observations, which are stored in the \texttt{Visit}

table. We distinguish it from a catalog as the data it contains was produced by the observatory directly, rather than the science pipelines.”

and

“Some of the information contained within the \texttt
{Visit}

table is also contained in the \texttt

{CCDVisit}

catalogue described in \secref
{ssec:catalogs}

(e.g., exposure time), although the latter also includes information produced by the processing pipelines at a per-detector level, such as the PSF size and limiting magnitudes."

>> In both Sections 3.2 and 3.5 clarify where all the useful data about an observation taken is stored.

We feel this is covered by addressing the previous point.

>> Section 3/Section 4 - It would be helpful to have normalized histograms per filter showing the magnitude distribution (flux distribution) of detected DP1 sources. One plot for the coadd source catalogs and one for the single epoch source catalogs.

We have added two normalized histograms showing the cumulative number of sources and objects as a function of PSF magnitude and band (Figure 9).

>> Limiting Magnitude/Depth - There's little discussion of the limiting magnitude/depth of this dataset particularly single epoch image processing/processed visit images (PVIs). I see there is a 5-sigma limiting magnitude in the CcdVisit table. There should be a histogram/visualization of the limiting magnitudes per filter to give the reader a sense of the depth of the

We have added a normalized histogram showing the cumulative number of visit_images as a function of 5sigma depth and band (Figure 8).

Comment by James Mullaney [10/Dec/25]

Both of these points have now been addressed.

Comment by James Mullaney [30/Jan/26]

You've already reviewed this and the changes have already been merged, so I'm just going to go straight to Done.

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